MACHINE LEARNING

ANALYSIS ON REAL WORLD DATA

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1. Unsupervised Learning

1.1 Problem statement

Committed to combating poverty and providing basic amenities and relief to victims of natural disasters and calamities, HELP International is an international humanitarian Non-Governmental Organisation (NGO). Having raised $10 Million, the CEO of the NGO must decide on the strategic and effective usage of these funds. Therefore, the CEO must identify the countries who are in the greatest need of aid.
This study aims to develop an unsupervised learning model to cluster the countries based on factors such as socio-economic and health. Analysing these clusters will assist in identifying the countries that need to be focused on the most.

1.2 Research Questions (RQ)

- **RQ1:** What is the optimal clusters for grouping countries?
- **RQ2:** What key socio-economic and public health factors distinguish the identified clusters of countries?
- **RQ3:** What are the implications of the clustering findings for future analysis?

1.3 Description of Dataset, and Exploratory Data Analysis (EDA)

The dataset comprises socio-economic and health-related indicators for 167 countries, with 10 features in total. There are no missing values in the dataset.

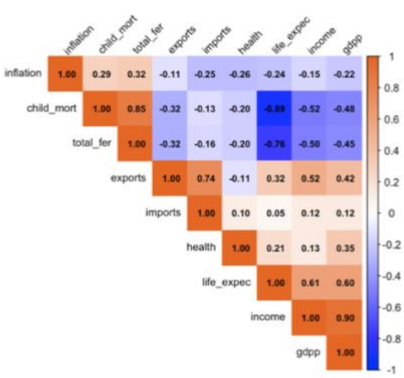

Figure 1.1 Correlation Matrix Heatmap

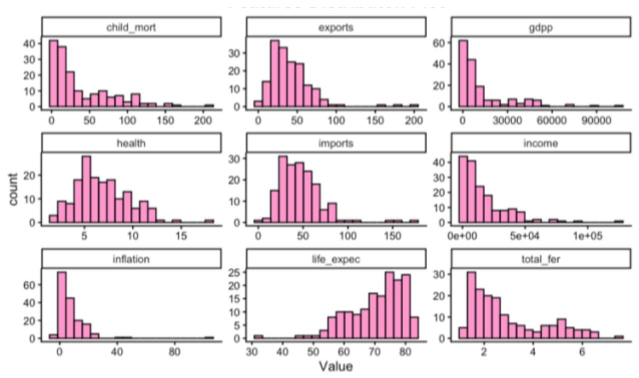

Figure 1.2 Features Distribution Plot

To examine the relationships between variables, a correlation heatmap (Figure 1.3.1) was constructed. From the heatmap, a strong positive correlation can be observed between income and GDP per capita (gdpp), with a coefficient of approximately 0.90. This indicates that countries with higher income levels generally have higher GDP per capita.

In addition, a strong negative correlation exists between child mortality and life expectancy, with a coefficient of around -0.89. This suggests that countries with higher rates of child mortality tend to experience lower life expectancy, likely reflecting poorer socio-economic and healthcare conditions.

The distribution plots (Figure 1.3.2) show that most variables, such as income and GDP per capita, are right-skewed. This indicates that a small number of countries have significantly higher values compared to the majority. In contrast, life expectancy appears to be slightly left-skewed, with most countries having relatively high life expectancy values.

Overall, these observations highlight the presence of strong relationships between economic and health indicators, and suggest the need for feature scaling before applying further analysis such as PCA and clustering.

1.4 Methodology

1.4.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied to the dataset as a technique to reduce the number of variables in a dataset, preserving as much original information as possible. This is done by transforming the original variables into a smaller set of linearly independent principal components, thereby simplifying the dataset and improving clustering performance.
A common way to determine the number of principal components to retain is by using a scree plot. It helps identify the point where the graph starts to flatten, indicating that additional components contribute less to explaining the data.

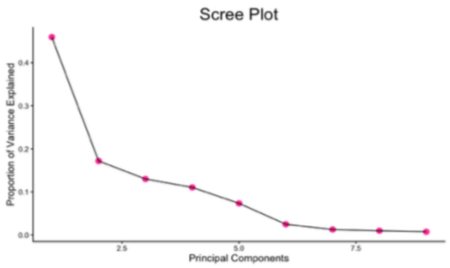

Figure 1.3 Scree plot

From the scree plot (Figure 1.3), there is a clear change in the slope after the second to third principal component, where the curve starts to flatten. This suggests that the first 2 to 3 components explain most of the variance, while the remaining components contribute very little. Hence, retaining 2-3 principal components is optimal.

1.4.2 K-Means clustering

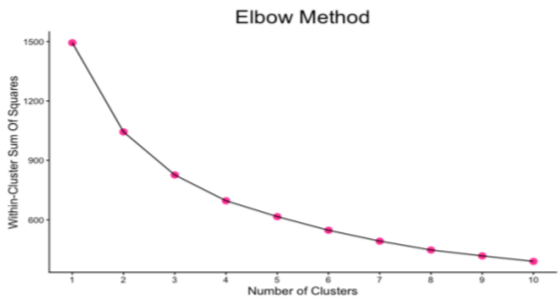

Figure 1.4 Elbow plot

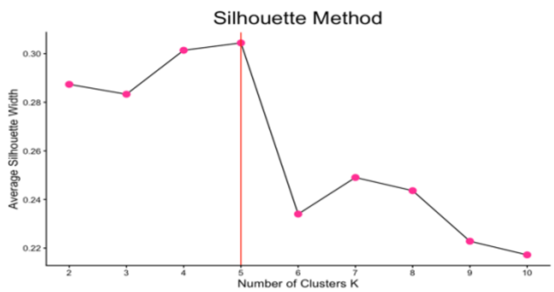

Figure 1.5 Silhouette plot

To determine the optimal number of clusters, the Elbow method was applied by plotting the within-cluster sum of squares (WSS) against k. As k increases, WSS decreases, but after a certain point, the rate of decrease slows down. This point is known as the “elbow” which indicates the optimal number of clusters. From the elbow plot (Figure 1.4), there is a clear bend around k = 3. Hence, 3 clusters were selected for the analysis

The Silhouette method was also used to evaluate how well each data point fits within its assigned cluster relative to other clusters. From Figure 1.5, we can observe that the highest score was at k = 5. However, k = 3 also provides a relatively high score and is consistent with the Elbow method and was thus selected for the analysis.

Boxplots (Figure 1.6) of different variables were created and analysed to further interpret the clusters.
Cluster 1 has the highest GDP, income, as well as higher life expectancy. It also has the lowest child mortality, suggesting more developed countries.
Cluster 3 shows the lowest GDP and income, with the highest child mortality, and lowest life expectancy, suggesting underdeveloped countries.
Lastly, cluster 2 lies between the previous two clusters, representing developing countries with moderate values across most variables.
Overall, the clusters show meaningful separation based on socio-economic and health indicators.

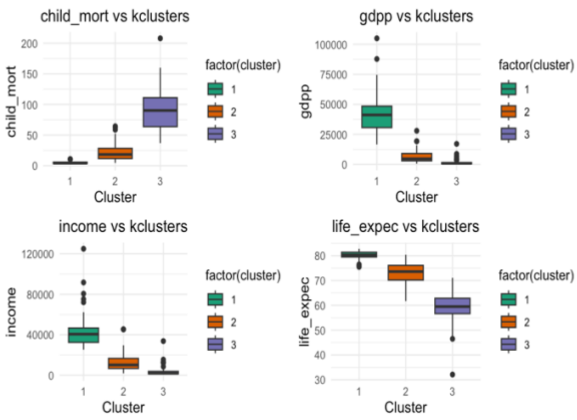

Figure 1.6 PCA Boxplots

1.4.3 Hierarchical Agglomerative Clustering

Hierarchical clustering is a method that groups data based on similarity. It builds a tree-like structure, called a dendrogram, which shows how data points are gradually merged into clusters. It is different from PCA as it focuses on grouping similar points into clusters, while PCA reduces the number of variables in a dataset.

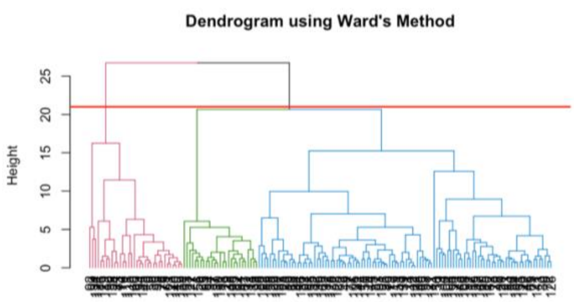

Figure 1.7 Dendrogram

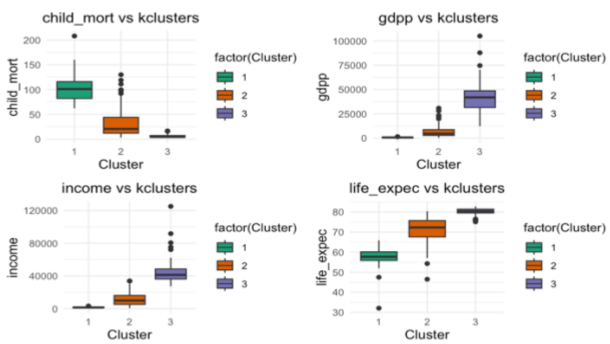

Figure 1.8 HCA Boxplots

A dendrogram (Figure 1.7) was created and cut at the chosen height, a clear separation into three clusters was observed, supporting the earlier findings from the Elbow and Silhouette method that the optimal number of clusters is three.
The clustering patterns observed in figure 1.8 are similar to that of figure 1.6, with the only difference being the cluster labels.

1.5 Conclusion

In conclusion, the results obtained from the clustering show that the optimal number of clusters for grouping countries is three, where they can be classified into three clusters: developed, developing, and underdeveloped.

Key factors such as GDP, income, and child mortality distinguish these clusters. Developed countries are characterised by higher income, life expectancy, and child mortality, while underdeveloped countries show the opposite. Developing countries fall between these two groups.

These categories highlight the correlation between economic performance and health outcomes, providing insight into global disparities. These findings can be used to help guide where and how aid should be allocated effectively by identifying countries that require the most help.

2. Regression

2.1 Problem statement

The life expectancy dataset by World Health Organisation (WHO) covers multiple countries over a period of time, along with related socio-economic and health indicators. These provide insight into the factors affecting living conditions and mortality.

This study aims to develop a regression model to analyse the relationship between these factors and life expectancy, helping to identify the key variables that have the greatest impact on life expectancy and how health outcomes can be improved.

2.2 Research questions

- **RQ1:** Which variables most significantly affect life expectancy?
- **RQ2:** Which regression model provides the best fit for the data?
- **RQ3:** What are the implications of the Regression findings for future research?

2.3 Dataset Description and Exploratory Data Analysis (EDA)

The dataset comprises socio-economic and health-related indicators for 193 countries, with 22 features in total. Missing values were checked for and handled using median imputation, ensuring no missing values before analysis.

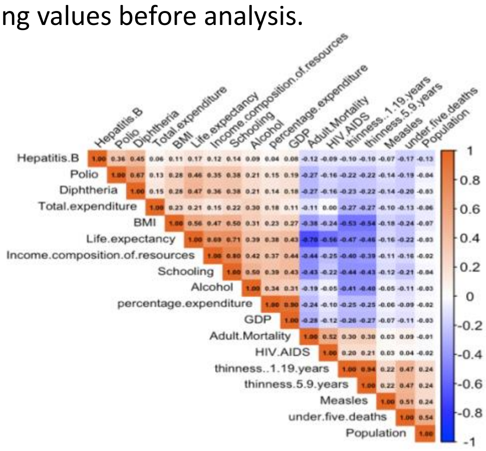

Figure 2.1 Correlation Heatmap

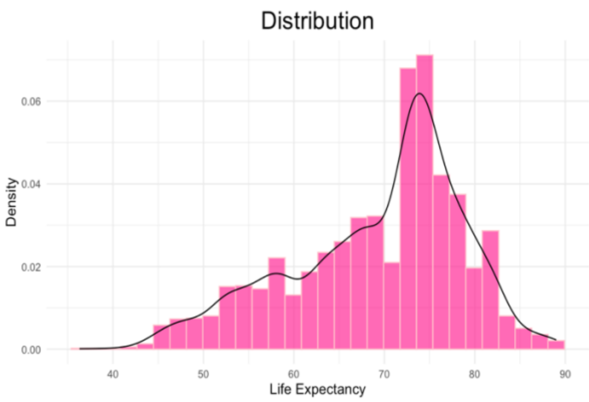

Figure 2.2 Distribution Plot

To examine the relationships between variables, a correlation heatmap (Figure 2.1) was created. A strong positive correlation between GDP and percentage expenditure can be observed, with a coefficient of approximately 0.90. This may suggest that higher economic is linked to increased spending. In contrast, there is a strong negative correlation between adult mortality and life expectancy (0.70), indicating that countries with higher adult mortality rates tend to have lower life expectancy. A distribution plot (Figure 2.2) was also plotted, and it shows that the plot is slightly left-skewed, with the average life expectancy concentrated between 65-80 years old.

Overall, strong relationships between the variables suggest that several features may be useful as predictors.

2.4 Methodology and Models

2.4.1 Linear and Polynomial Regression

Linear regression is a technique used to model the relationship between variables by fitting a best-fit line to the data. It predicts continuous outcomes by minimising the difference between observed and predicted values. In contrast, polynomial regression fits a curve instead of a straight line, enabling it to model non-linear relationships within the data.

Both linear (Figure 2.3) and polynomial (Figure 2.4) regressions were plotted, comparing between the predicted and actual values. Both plots show a strong positive relationship. The predictions follow the diagonal line closely, which suggests that the model has good overall performance. However, the polynomial regression captured the spread of the data better, especially at higher values, indicating that it may handle non-linear patterns more effectively.

The RMSE (Root Mean Squared Error) was also calculated. RMSE represents the average difference between predicted and actual values, with lower values indicating better model performance. The polynomial regression has a slightly lower RMSE at 3.96 compared to the linear regression with 3.97. However, as the difference is minimal, it means that a linear model is sufficient, as both models performed similarly.

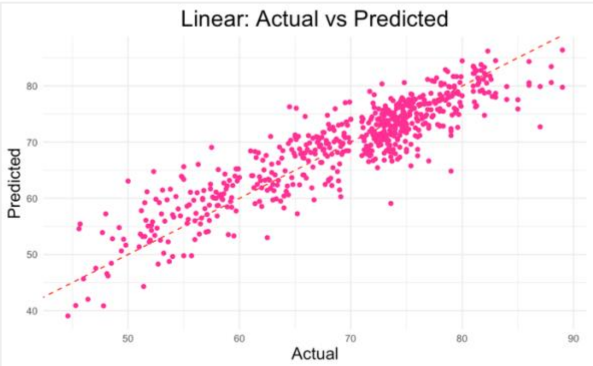

Figure 2.3 Linear Regression

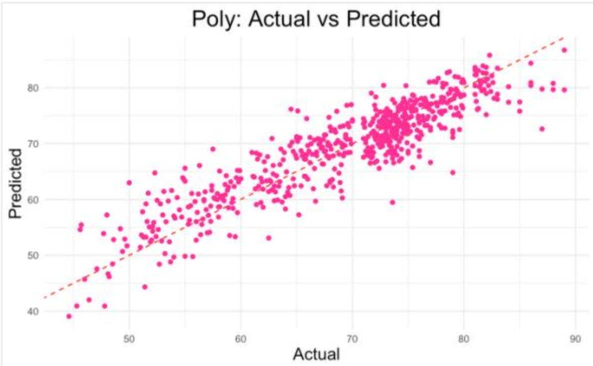

Figure 2.4 Polynomial Regression

2.4.2 Regularised Regression (Ridge and Lasso)

Regularised regression is used to prevent overfitting, which improves model performance. Ridge and Lasso are regularised linear regression models, where Ridge shrinks coefficients towards zero without eliminating them, while Lasso performs automatic feature selection by setting some coefficients to zero.

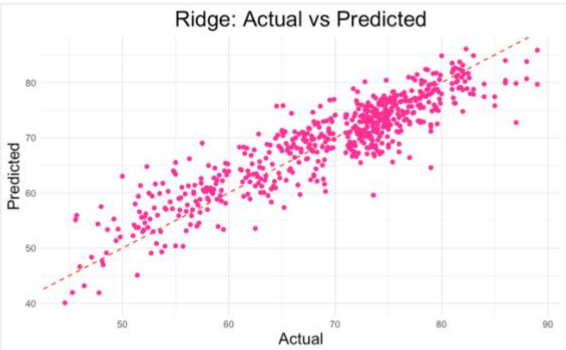

Figure 2.5 Ridge Regularised Regression

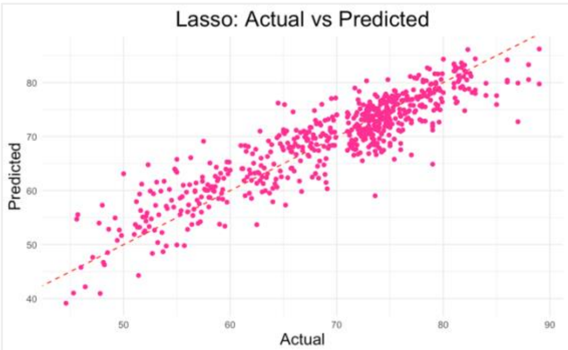

Figure 2.6 Lasso Regularised Regression

Similarly, both Ridge (Figure 2.5) and Lasso (Figure 2.6) regression models were created and compared.
Both models show a strong positive relationship between actual and predicted values, with points closely following the diagonal line, suggesting good overall model performance. There is no clear visual difference between the two models, as both capture the pattern of the data similarly.

The calculated RMSE for the Ridge regression was 3.971, slightly higher than the Lasso regression RMSE, which was 3.966, meaning that the Lasso performed slightly better in the prediction of life expectancy. Overall, the difference in RMSE is minimal, indicating that both models perform similarly, with Lasso providing only a slight improvement

A possible explanation for Lasso's better performance is its ability y to perform feature selection by shrinking some coefficients to zero, eliminating less important variables may be the reason for its marginally better performance. Ridge regression, on the other hand, keeps all the variables but reduces their influence.

2.4.3 Random Forest

Lastly, a Random Forest model was created and plotted to visualise.
Random Forest is an ensemble learning method used for classification and regression, which operates by constructing multiple decision trees during training. For regression tasks, the final prediction is obtained by averaging the outputs of all the trees.

As shown from Figure 2.7, the points were closely aligned along the diagonal line, which indicates strong predictive accuracy. Compared to the previous models, the predictions seem more tightly clustered. This suggests that the Random Forest captures the relationships in the data more effectively and accurately.
The calculated RMSE was 1.88 for the Random Forest model, which makes it the lowest amongst all models.

The contribution of each variable to the model’s predictions is shown in a feature importance plot (Figure 2.8). The plot shows that HIV/AIDS is the most influential variable, followed by income composition of resources and adult mortality.

Due to its accuracy and ability to capture more complex relationships in the data, the Random Forest model performs better than the rest of the models, as evident by its low RMSE.

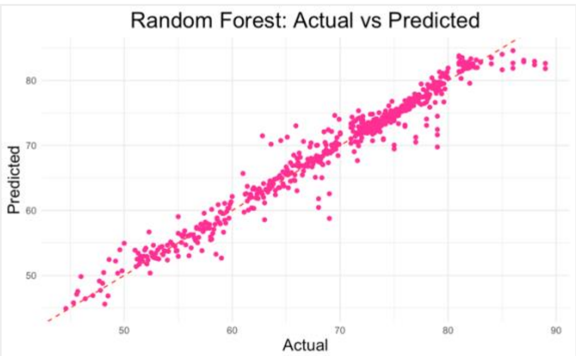

Figure 2.7 Random Forest

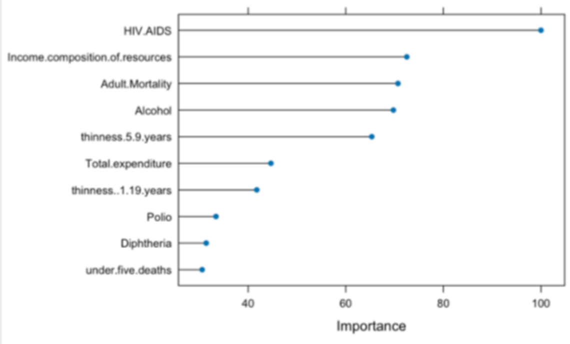

Figure 2.8 Feature Importance

2.5 Conclusion

In conclusion, variables such as HIV/AIDS, income composition of resources, and adult mortality have been identified to have the most significant impact on life expectancy. This is supported by the results shown in the feature importance plot, highlighting the importance of health-related and socio-economic factors in determining overall population health.

The model which performed the best is the Random Forest model, which provided the best fit. It had the lowest RMSE compared to all the models, indicating that it is accurate.

The findings suggest that key factors such as diseases prevalence and economic conditions should be highly focused on due to their strong influence on life expectancy.

3. Classification

3.1 Problem Statement

Millions in the United States affected by diabetes, a prevalent chronic disease which imposes a significant health and economic burden. Opportunities for early intervention are limited due to a large portion of individuals who suffer from diabetes or prediabetes remaining undiagnosed. There is a critical need for effective early detection to identify individuals that are at risk.

The aim of this study is to help identify diabetes using classification models that predict whether an individual is at risk based on relevant health and demographic factors.

3.2 Research questions

• **RQ1:** Which features most significantly influence the classification of individuals as diabetic or non-diabetic?
• **RQ2:** Which classification model achieves the highest predictive performance for diabetes detection?
• **RQ3:** What are the implications of the Regression findings for future research?

3.3 Dataset Description and Exploratory Data Analysis (EDA)

The dataset consists of 22 variables, and 21 predictor features covering health conditions, lifestyle factors, and demographic information. Missing and duplicated values were checked for and omitted.

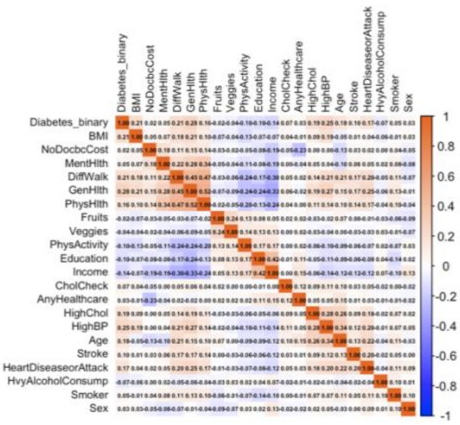

Figure 3.1 Correlation Heatmap

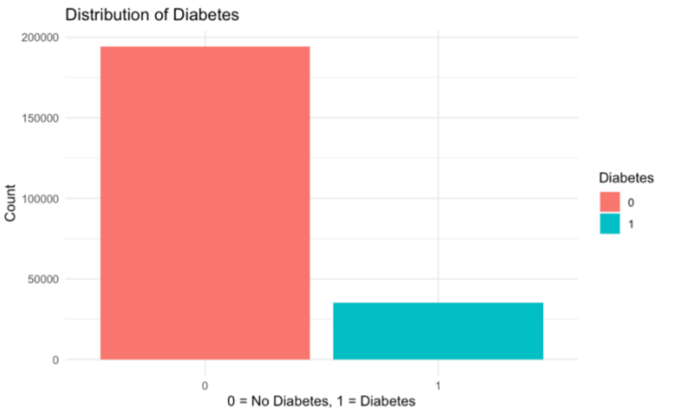

Figure 3.2 Distribution of diabetes

To examine the relationships between variables, a correlation heatmap (Figure 3.1) was created. The results show generally weak to moderate correlations between most variables and the target variables, with the strongest positive correlations observed between diabetes, and GenHlth (0.28). This suggests that poorer general health is associated with a higher likelihood of diabetes. In contrast, most other variables exhibit very weak or near-zero correlations with the target, indicating limited linear relationships. Overall, there is no single variable identified to strongly determine diabetes and is instead likely influenced by many other factors.

There is a clear class imbalance as seen by the distribution (Figure 3.2), with 85% being classified as non-diabetic, while 15% as diabetic. This imbalance may bias models towards the majority class and hence, making factors such as balanced accuracy and ROC-AUC more appropriate for evaluation than accuracy alone.

3.4 Methodology and Models

3.4.1 Logistic Regression

Logistic regression is a supervised machine learning algorithm, which is used to analyse the relationship between predictor variables and a categorical outcome. Logistic regression is used for binary outcomes, like whether an event occurs or not, unlike linear regression, which predicts the likelihood that an input belongs to a particular class.

The accuracy of the logistic regression model is approximately 84.9%, which means that the model is able to accurately and correctly classify majority of the observations. However, due to the class imbalance in the dataset, accuracy alone is not enough to evaluate the performance of the model.

It also scored a high F1 score of 0.916, indicating that it is effective in correctly identifying whether an individual is diabetic or not, while also minimising false positives and negatives.

Lastly, the ROC-AUC value of the model was 0.805, reflecting the good discriminatory ability. This shows that the model is able to distinguish diabetic individuals from non-diabetic individuals effectively across different classification thresholds.

3.4.2 Random Forest (RF)

Random Forest is a machine learning algorithm. By merging the outputs of multiple decision trees, it improves prediction accuracy. Different random subsets of the data are used to train each tree, allowing them to learn diverse patterns. The results from all trees are combined to make the final prediction.

The accuracy of the Random Forest model is approximately 85.2%, which indicates that the model is able to correctly classify a large proportion of the observations.

The model achieved a high F1 score of 0.918, indicating that it performs very well in identifying diabetic individuals while maintaining a good balance between precision and recall.

Lastly, the ROC-AUC value of the model was 0.787 meaning good discriminatory ability.

3.4.3 K-Nearest Neighbour (K-NN)

K-Nearest Neighbours (K-NN) is a supervised learning algorithm. It identifies the k closest data points to a given observation and makes predictions based on those neighbours. For classification, the model assigns the most common class among the nearest neighbours.

The accuracy of the KNN model is approximately 83.1%, which suggests that the model is able to correctly classify a reasonable proportion of the observations.

The model achieved a very low F1 score of 0.271, indicating poor performance in correctly identifying diabetic individuals, implying that the model had trouble striking balance between recall and position.

Finally, the K-NN model's ROC-AUC value was 0.708, suggesting comparatively poor discriminatory ability.

3.4.4 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification It works by determining the best boundary (also called a hyperplane) that separates various classes of data. In order to improve the accuracy on new data, SVM chooses the boundary that maximises the margin (distance between the line and closest data points).

The SVM model achieved an accuracy of approximately 84.7%, meaning that the model is able to correctly classify a large proportion of the observations.

The model has a very low F1 score of 0.133, indicating extremely poor performance in correctly identifying diabetic individuals.

Lastly, the ROC-AUC value that the SVM model obtained was 0.721, highlighting the poor performance of the model in telling diabetic and non-diabetic individuals apart.

3.5 Area under the Receiver Operating Characteristic Curve

The Receiver Operating Characteristic (ROC) – Area Under Curve (AUC) is used to evaluate the performance of a binary classification model. It demonstrates the effectiveness of the model in distinguishing between two classes, in this case, diabetic and non-diabetic individuals, across varied threshold values. Multiple curves were created and plotted (Figure 3.3). The AUC value represents the overall performance of the model.

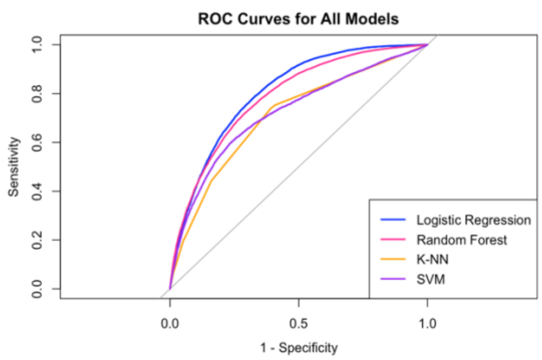

Figure 3.3 ROC Curve for all models

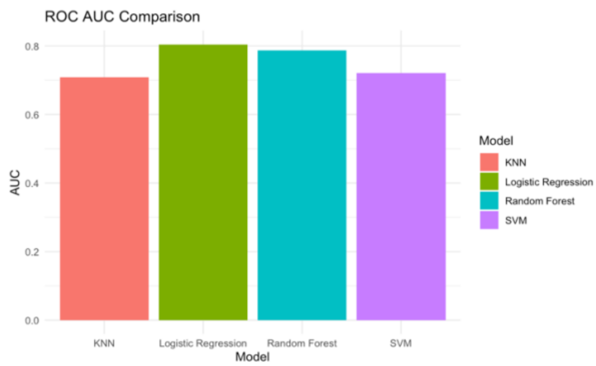

Figure 3.4 ROC comparison overall

As their curves lie above the diagonal line, the ROC curves (Figure 3.3) show that all models perform better than random guessing. This is also shown in Figure 3.4. With its curve consistently above the rest, Logistic Regression demonstrated the strongest performance, indicating that it does the best in differentiating between diabetic and non-diabetic individuals across different thresholds.
Random Forest also performed well, with a curve very close to Logistic Regression, suggesting similarly strong discriminatory ability.

However, in contrast, K-NN and SVM show weaker performance, as their curves are further from the top-left corner, indicating lower true positive rates. Overall, Logistic Regression exhibits the highest discriminatory power, followed closely by Random Forest, while KNN and SVM perform less effectively. This means that Logistic Regression may be the most reliable model due to its discriminatory performance.

3.6 F1 and Accuracy

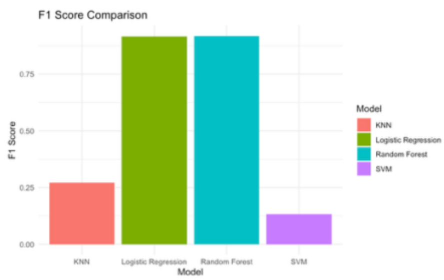

Figure 3.5 F1 Comparison

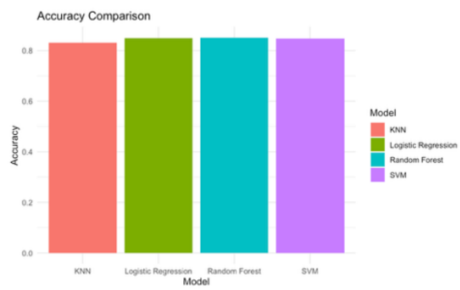

Figure 3.6 F1 Comparison

The F1 score is a metric that provides a balanced measure of a model's accuracy.

The F1 score and accuracy comparisons show clear differences in model performance. Logistic Regression and Random Forest achieved the highest F1 scores, both around 0.91, indicating strong ability to correctly identify diabetic individuals while maintaining a good balance between precision and recall. In contrast, KNN recorded a much lower F1 score of approximately 0.27, while SVM performed the worst with an F1 score of around 0.13, suggesting poor detection of the minority class.

In terms of accuracy, all models achieved relatively similar results, ranging from approximately 83% to 85%. Random Forest had the highest accuracy, followed closely by Logistic Regression and SVM, while KNN had the lowest. However, due to the class imbalance in the dataset, accuracy alone is not a reliable indicator of performance.

3.6 Conclusion

In conclusion, the results show that diabetes is influenced by a combination of health, lifestyle, and demographic factors rather than a single dominant feature. In terms of model performance, Random Forest achieved the best overall results, with the highest accuracy and F1 score, while Logistic Regression also performed well, particularly in ROC-AUC. In contrast, KNN and SVM showed weaker performance, especially in identifying cases of diabetes.

Overall, the findings suggest that ensemble models such as Random Forest are more suitable for this task, and that evaluation metrics like F1 score and ROC-AUC are more appropriate than accuracy when dealing with imbalanced datasets

References

1. GeeksforGeeks. (2026, March 18). *Hierarchical clustering in machine learning*. https://www.geeksforgeeks.org/machine-learning/hierarchical-clustering/
2. Hanson, B. A., & Harvey, D. T. (2024, April 25). *Step-by-step PCA* (LearnPCA package vignette, Version 0.3.4). CRAN. https://cran.r-project.org/web/packages/LearnPCA/vignettes/Vig_03_Step_By_Step_PCA.pdf
3. IBM. (n.d.). *What is logistic regression?* https://www.ibm.com/think/topics/logistic-regression
4. IBM. (n.d.). *What is a support vector machine (SVM)?* https://www.ibm.com/think/topics/support-vector-machine
5. ScienceDirect. (n.d.). *Scree plot*. Elsevier. https://www.sciencedirect.com/topics/mathematics/scree-plot
6. Statology. (2022, September 8). *How to use the elbow method in R to find optimal clusters*. https://www.statology.org/elbow-method-in-r/
7. UC Business Analytics R Programming Guide. (n.d.). *K-means clustering*. https://uc-r.github.io/kmeans_clustering